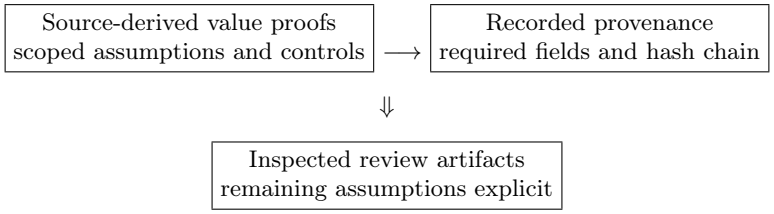
Native bridges and normalized output validity

This continuation completes three scoped proof components and strengthens the saved-file accounting checker. The packet records actual assumptions and failed attempts separately from passing checks. No scientific campaign or target enumeration was run.



C-zero output: twelve main declarations and ten controls establish the normalized output curve equation from the original input divisibility hypotheses. For monic quintic f with coefficient four zero, the residual divided by the two quadratic supports has degree one. Its coefficients give X-x, where x=q1+q2+(A/B) squared. With l=v1-(A/B)u1, the normalized candidate ordinate y=-l(x) satisfies y squared=f(x). The quotient is derived rather than assumed. Weighted scale transport and divisor-class semantics remain to connect.

Native graph arithmetic: seven main declarations and ten producer controls cover literal add, negate, subtract, widened multiply and the signed branch. Independent review added nine controls with twenty-six axiom readbacks. Canonical add/sub use r at most 2^63; the native constructor does not enforce that bound. Multiplication widens before multiplying and reduces before narrowing. Sign representation, inversion, mutable graph invariants and counter bounds remain distinct.

Kummer comparator: six main declarations and eight producer controls prove that the sixteen cross-product comparisons characterize nonzero scalar proportionality for nonzero field vectors. Independent review added six controls with twenty readbacks. The raw comparator matches zero to every vector; controls demonstrate failed transitivity when zero is admitted. Canonical word decoding and concrete projection fibers remain separate source bridges.

Accounting v2: required observed/index identity fields and index labels are compared to the archive. The recorded receipt-to-manifest-to-archive/index hashes and byte lengths now verify. Accounting and provenance pass flags remain separate. Five producer negative controls fail closed. Root rebuilt the checker separately, reproduced byte-identical positive output, and added two failing controls for a wrong receipt hash and a wrong archive hash in a rehashed manifest. All original inputs remain unchanged. This checks recorded consistency, not receipt authenticity or output execution.

The saved accounting partition remains 180 physical rows, 60 selected rows and 420 dispositions. Full cohort scope remains 1,260 targets; full-cost ratios and natural yield remain null. The campaign deadline remains 2026-10-31 06:00 UTC. Further scientific execution requires renewed limits and the recorded qualification/review checks.

C-zero independent replay passed twelve main declarations, ten producer controls and five reviewer declarations, with twenty-seven axiom readbacks. It additionally checked the literal unit-input-weight ordinate normalization Y/B^5=-l(X/B^2), with B nonzero. Other independent reviews are scoped technical checks, with broader workflow and compiled-execution boundaries recorded. Failed producer and reviewer attempts remain in the packet. The shared session-log append still awaits the existing reservation exception.